MEDIUM · ADVANCED MATH

Advanced Math

18

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

For the exponential function f, the value of $f0$ is c, where c is a constant. Of the following equations that define the function f, which equation shows the value of c as the coefficient or the base?

A. $fx = 221.5^{x+1}$

B. $fx = 331.5^{x}$

C. $fx = 49.51.5^{x-1}$

D. $fx = 74.251.5^{x-2}$

Q2

ADVANCED MATH

$$S(n) = 38{,}000a^n$$

The function S above models the annual salary, in dollars, of an employee n years after starting a job, where a is a constant. If the employee's salary increases by 4% each year, what is the value of a ?

A. 0.04

B. 0.4

C. 1.04

D. 1.4

Q3 ADVANCED MATH

A scientist initially measures 12,000 bacteria in a growth medium. 4 hours later, the scientist measures 24,000 bacteria. Assuming exponential growth, the formula $P = C2^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. What is the value of r?

A. $\frac{1}{12{,}000}$

B. $\frac{1}{4}$

C. 4

D. 12,000

Q4 ADVANCED MATH

A model predicts that the population of Bergen was 15,000 in 2005. The model also predicts that each year for the next 5 years, the population p increased by 4% of the previous year's population. Which equation best represents this model, where x is the number of years after 2005, for $x \leq 5$?

A. $p = 0.9615{,}000^x$

B. $p = 1.0415{,}000^x$

C. $p = 15{,}0000.96^x$

D. $p = 15{,}0001.04^x$

Q5 ADVANCED MATH

The function f is defined by $fx = 7x^3$. In the xy-plane, the graph of $y = gx$ is the result of shifting the graph of $y = fx$ down 2 units. Which equation defines function g?

A. $gx = \frac{7}{2}x^3$

B. $gx = 7x^{\frac{3}{2}}$

C. $gx = 7x^3 + 2$

D. $gx = 7x^3$ - 2

Q6 GRID-IN ADVANCED MATH

$$fx = x + 6x - 4$$

If the given function f is graphed in the xy-plane, where $y = fx$, what is the x-coordinate of an x-intercept of the graph?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7 ADVANCED MATH

$$q(x) = 32(2^x)$$

Which table gives three values of x and their corresponding values of $q(x)$ for function q?

A.

x	-1	0	1
$q(x)$	-64	0	64

B.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	2	64

C.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	32	64

D.

x	-1	0	1
$q(x)$	16	32	64

Q8 ADVANCED MATH

An object's kinetic energy, in joules, is equal to the product of one-half the object's mass, in kilograms, and the square of the object's speed, in meters per second. What is the speed, in meters per second, of an object with a mass of 4 kilograms and kinetic energy of 18 joules?

A. 3

B. 6

C. 9

D. 36

Q9 ADVANCED MATH

The area A, in square centimeters, of a rectangular painting can be represented by the expression $ww + 29$, where w is the width, in centimeters, of the painting. Which expression represents the length, in centimeters, of the painting?

A. w

B. 29

C. $w + 29$

D. $ww + 29$

Q10 ADVANCED MATH

A company has a newsletter. In January 2018, there were 1,300 customers subscribed to the newsletter. For the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. Which equation gives the total number of customers, c, subscribed to the company's newsletter m months after January 2018, where $m \leq 24$?

A. $c = 1{,}3000.07^{m}$

B. $c = 1{,}3001.07^{m}$

C. $c = 1{,}3001.7^{m}$

D. $c = 1{,}3007^{m}$